Omar Ibrahim

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Education

Carleton University Ottawa, ON
Bachelor of Engineering in Software Engineering *Apr. 2028*

Technical Skills

Programming Languages: Python, C++, TypeScript, JavaScript, Java, SQL, HTML/CSS, Bash Scripting
Web & Backend: Next.js, React, Node.js, REST APIs, Tailwind CSS
Databases: PostgreSQL, Supabase
Infrastructure & Systems: Linux, Windows, VPS Hosting, Docker, CI/CD, GitHub Actions, Cloudflare
Networking & Security: SSH, Secure Remote Access, Firewall Configuration, Access Control Auditing, Web Penetration Testing

Experience

Cybersecurity Analyst Remote
Journale AI *Oct. 2025 – Dec. 2025*

- Conducted penetration testing across **3 core attack surfaces** (authentication, payments, APIs), identifying authorization and data exposure vulnerabilities.
- Reproduced exploits and collaborated with engineers to deploy secure remediations under responsible disclosure.
- Assessed access control failures including IDOR, privilege escalation paths, and session handling weaknesses.
- Advised the development team on secure deployment and operational practices, helping transition the site to a more resilient configuration suitable for an organization handling public content and donations.

Freelance Web Developer Ottawa, ON
Self-Employed *Sep. 2025 – Present*

- Designed, deployed, and maintained production websites for **multiple clients**, handling full-stack development and infrastructure ownership.
- Managed Linux VPS environments including domain configuration, SSL, firewall hardening, backups, and uptime monitoring.
- Implemented CI/CD pipelines to automate deployments and reduce manual release overhead.

Co-President & Web Developer Ottawa, ON
Empower Orphans Foundation *Mar. 2024 – Present*

- Co-founded and led a **50+ member** organization across marketing, events, sponsorship, and internal operations.
- Built and deployed a full-stack platform using Next.js and Supabase with role-based admin access and secure data handling.
- Directed fundraising initiatives and partnerships supporting orphan care through Islamic Relief collaborations.

Projects

CUMSA Security Auditor | Web Security, Access Control, Linux 2025

- Conducted an authorized security audit of a student-facing web platform supporting **2,000+ students**, identifying and reporting multiple high-impact vulnerabilities including broken access control, credential exposure, path traversal, unsafe file handling, and missing security headers.
- Documented step-by-step reproduction and remediation guidance for **8+ distinct security issues**, enabling rapid fixes and preventing unauthorized access, data exposure, and content manipulation.
- Hardened authentication and API authorization flows, eliminating unauthenticated event creation/deletion and reducing the site's effective attack surface by **multiple orders of magnitude**.

CUMSA Discord Bot Contributor | Node.js, Discord.js v14, Git 2025

- Migrated legacy bot commands to modern slash commands while following structured Git workflows and formal pull request reviews. Improved maintainability and consistency across command handling and permissions logic.

University of Ottawa MSA Platform | React, Supabase, PostgreSQL 2025

- Collaborated within an **8-developer** engineering team to rebuild a platform supporting **1,000+ students**.
- Redesigned UI components, refactored database schemas, and developed secure admin dashboard features.
- Worked in Agile sprints using GitHub pull requests, CI/CD pipelines, and structured code reviews.

Secure Home Lab & Cloud Infrastructure | Linux, Windows, VPS, Docker, CI/CD 2025

- Designed and deployed a hybrid infrastructure spanning a Linux VPS and **3 private Windows/Linux nodes** for remote development and compute workloads.
- Implemented zero-trust remote access using encrypted overlay networking, resulting in **zero public inbound ports**.
- Hardened headless systems with full-disk encryption, firewall-scoped access rules, and least-privilege authentication.